

WATER WASTAGE REPORTING PLATFORM

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Abstract

Water is one of the most important natural resources for human survival, agriculture, industry, and ecosystem sustainability. However, a significant amount of water is wasted every day due to leaking pipelines, overflowing water tanks, damaged public taps, and irresponsible usage practices. In many cases, water wastage continues for long periods because authorities are not informed promptly. This report proposes a Water Wastage Reporting Platform, an information technology solution that enables citizens to report incidents of water wastage using a web or mobile application. The platform allows users to upload photographs, provide location details, and submit complaints directly to the responsible authorities. The authorities can monitor reports, assign corrective actions, and update the status of complaints. The proposed system promotes community participation, improves response time, and contributes to sustainable water management. The solution supports several Sustainable Development Goals and can help reduce unnecessary water losses in urban and rural areas.

1. Introduction

Water is essential for life and economic development. Despite its importance, freshwater resources are becoming increasingly scarce due to population growth, urbanization, climate change, and excessive consumption. One major contributor to water scarcity is water wastage.

Water wastage occurs when clean water is lost through leaks, overflowing tanks, damaged infrastructure, or careless consumption practices. In many cities and villages, leaking pipelines may waste thousands of liters of water every day. Often these problems remain unresolved because the responsible authorities are not informed in time.

Modern information technology provides opportunities to involve citizens in water conservation efforts. Through digital platforms, people can quickly report issues and help authorities identify areas where water is being wasted. Therefore, an efficient reporting mechanism can play a significant role in reducing water losses and promoting sustainable resource management.

This report proposes a Water Wastage Reporting Platform that connects citizens and authorities through a centralized digital system.

2. Problem Statement

Water wastage -water conservation monitoring portal

Currently, many water wastage incidents are reported through manual methods such as phone calls or office visits. These methods may be slow, inefficient, and difficult to track. As a result, water continues to be wasted before corrective actions are taken.

There is a need for an efficient, transparent, and user-friendly system that enables citizens to report water wastage incidents and allows authorities to monitor and resolve them effectively.

3. Sustainability Analysis

3.1 Relevant Sustainable Development Goals (SDGs)

SDG 6: Clean Water and Sanitation

The proposed solution directly contributes to SDG 6 by reducing water losses and encouraging efficient water management.

SDG 11: Sustainable Cities and Communities

The platform supports sustainable urban infrastructure by helping authorities identify and repair water distribution issues.

SDG 12: Responsible Consumption and Production

The system promotes responsible water usage and reduces unnecessary wastage.

SDG 13: Climate Action

Efficient water management reduces pressure on water resources and improves resilience against climate-related water shortages.

3.2 ISO 14001

ISO 14001 focuses on environmental management systems and continuous improvement.

The proposed platform supports ISO 14001 principles by:

- Identifying environmental issues.
- Facilitating corrective actions.
- Monitoring environmental performance.
- Supporting resource conservation initiatives.

3.3 Environmental Impact Assessment (EIA)

The environmental impacts of implementing the platform are largely positive.

Positive impacts include:

- Reduced water wastage.
- Improved conservation awareness.
- Faster repair of damaged infrastructure.
- Better management of water resources.

Negative impacts are minimal and mainly involve energy consumption by servers and digital infrastructure.

Overall, the environmental benefits significantly outweigh the environmental costs.

4. Proposed Engineering/IT Solution

Water Wastage Reporting Platform

The proposed solution is a web and mobile-based platform that allows citizens to report water wastage incidents directly to local authorities.

Main Features

User Registration

Citizens can create accounts using their email addresses or mobile numbers.

Report Submission

Users can submit complaints by providing:

- Description of the issue.
- Photographs of water wastage.
- GPS location.

- Date and time.

Location Tracking

The platform automatically records the location of the reported incident, helping authorities identify the exact area.

Complaint Management Dashboard

Authorities can access a dashboard displaying:

- New complaints.
- Pending complaints.
- Resolved complaints.
- Geographic distribution of incidents.

Status Updates

Authorities can update complaint status as:

- Received.
- Under Inspection.
- In Progress.
- Resolved.

Notifications

Users receive notifications whenever the status of their complaint changes.

Analytics Module

The system generates reports showing:

- Most affected locations.
- Frequency of incidents.
- Response times.
- Resolution rates.

System Architecture

- User reports water wastage through mobile app or website.
- Data is stored in a central database.
- Authorities receive notifications.
- Maintenance teams investigate the issue.
- Corrective actions are performed.
- Status is updated and communicated to users.

5. Feasibility Analysis

Technical Feasibility

The required technologies already exist and are widely used.

Required technologies include:

- Mobile application development.
- Web development.
- Cloud databases.
- GPS integration.
- Notification systems.

Therefore, technical feasibility is high.

Economic Feasibility

The platform can be developed using open-source technologies, reducing implementation costs.

Major expenses include:

- Development costs.
- Server maintenance.
- Training personnel.

Compared to the value of water saved, the costs are reasonable.

Operational Feasibility

Citizens are familiar with smartphones and mobile applications, making the system easy to adopt.

Government departments can integrate the platform into existing workflows with minimal changes.

Social Feasibility

The platform encourages public participation in environmental protection and is likely to receive positive community support.

6. Expected Environmental Benefits

The proposed platform can provide several environmental benefits:

- Reduction in water wastage.
- Conservation of freshwater resources.
- Improved infrastructure maintenance.
- Increased public awareness regarding water conservation.
- Faster detection of water distribution problems.
- Reduced pressure on reservoirs and groundwater sources.
- Better sustainability of urban water systems.

Over time, even small reductions in daily water wastage can result in significant water savings.

7. Justification of the Proposed Solution

The Water Wastage Reporting Platform is justified because it addresses one of the main causes of water wastage: delayed identification and reporting of problems.

The solution is:

- Cost-effective.
- Easy to implement.
- Scalable.
- Environmentally beneficial.
- Suitable for both urban and rural areas.

Unlike traditional reporting methods, the platform provides transparency and accountability. Citizens can actively participate in water conservation efforts, while authorities gain access to real-time information.

The system creates a collaborative approach to solving water wastage issues and supports sustainable development objectives.

8. Conclusion

Water wastage is a serious challenge that threatens environmental sustainability and future water security. Leaking pipelines, overflowing tanks, and damaged infrastructure contribute significantly to the loss of valuable freshwater resources. The proposed Water Wastage Reporting Platform provides a practical and efficient information technology solution for addressing this issue.

By enabling citizens to report incidents and allowing authorities to monitor and resolve complaints effectively, the platform can reduce water losses, improve response times, and promote responsible resource management. The solution supports Sustainable Development Goals, aligns with environmental management principles, and offers long-term benefits for communities and ecosystems.

Therefore, the implementation of a Water Wastage Reporting Platform can play an important role in achieving sustainable water conservation and environmental protection.

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